LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Daily Practice: Ch 2 Acids, Bases and Salts

Chapter / Topic: Acids, Bases and Salts · Week 1 of the 90-day plan

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DAY 1: ACIDS, BASES AND INDICATORS

Q1. Which of the following is an olfactory indicator?

Answer: (b) clove oil: its smell disappears in a basic solution (vanilla and onion work the same way)

Q2. Which acid is present in tamarind?

Answer: (d) tartaric acid

Q3. In water, acids produce:

Answer: (b) H^+ ions, which join water molecules to form H_3O^+

Q4. A base that dissolves in water is called:

Answer: (d) an alkali, e.g. NaOH, KOH

Q5. Name the acid present in: (i) vinegar (ii) curd (iii) an ant sting (iv) lemon.

Answer: (i) acetic (ethanoic) acid (ii) lactic acid (iii) methanoic (formic) acid (iv) citric acid

Q6. HCl and H_2SO_4 show acidic properties in water, but solutions of glucose and alcohol do not, although they also contain hydrogen. Why?

Answer: Acids give H^+ ions when dissolved in water, and these ions cause acidic behaviour. Glucose and alcohol do not split into ions in water, so they produce no H^+ ions.

Q7. Zinc reacts with sodium hydroxide solution on warming. Write the balanced equation and name the gas formed.

Answer: $2NaOH + Zn \rightarrow Na_2ZnO_2 + H_2$. The gas is hydrogen; the salt is sodium zincate.

Q8. You have three test tubes. One contains distilled water, one an acid and one a base. You have only red litmus paper. How will you identify each one?

Answer: Dip red litmus into each: the one that turns it blue is the base. Now use this blue litmus strip on the other two: the one that turns it red again is the acid. The one with no effect is distilled water.

DAY 2: REACTIONS AND PH

Q9. Metal oxides are generally:

Answer: (c) basic: they react with acids to give a salt and water

Q10. The pH of the gastric juice in our stomach is about:

Answer: (a) 2: the stomach makes hydrochloric acid, which helps digestion

Q11. Solutions of equal concentration are made of each substance below. Which has the lowest pH?

Answer: (a) HCl: it is a strong acid, so it gives the most H^+ ions

Q12. Write the balanced equations for the reactions of dilute hydrochloric acid with (i) copper(II) oxide and (ii) magnesium hydroxide. What do both…

Answer: (i) $CuO + 2HCl \rightarrow CuCl_2 + H_2O$ (the solution turns blue-green)
(ii) $Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$
Both give a salt and water: the oxide and the hydroxide behave as bases.

Q13. Why does an ant sting cause pain and a burning feeling? What can be applied to the skin for relief, and why does it work?

Answer: The ant injects methanoic (formic) acid into the skin. Rubbing on a mild base such as baking soda solution neutralises the acid and relieves the pain.

Q14. Fresh milk has a pH of about 6. (a) How does the pH change as it turns into curd? Why? (b) A milkman adds a very small amount of baking soda to fresh…

Answer: (a) The pH falls, because lactic acid forms and the milk becomes more acidic.
(b) Slightly alkaline milk does not turn sour as quickly, so it stays fresh for longer.
(c) The first lactic acid that forms is neutralised by the baking soda, so it takes longer for the milk to become acidic enough to set.

Q15. Salt solutions can be acidic, basic or neutral. Classify each with a reason: (i) NaCl (ii) Na_2CO_3 (iii) NH_4Cl.

Answer: (i) NaCl: neutral (pH 7), from a strong acid (HCl) and a strong base (NaOH).
(ii) Na_2CO_3: basic (pH > 7), from a strong base (NaOH) and a weak acid (H_2CO_3).
(iii) NH_4Cl: acidic (pH < 7), from a strong acid (HCl) and a weak base (NH_4OH).

DAY 3: SALTS, MIXED PRACTICE AND CHALLENGE

Q16. **Assertion (A):** Plaster of Paris sets into a hard mass when it is mixed with water. **Reason (R):** Plaster of Paris contains no water of crystallisation.

Answer: (c) A is true, but R is false: Plaster of Paris is $CaSO_4 \cdot \frac{1}{2}H_2O$, so it has half a molecule of water per $CaSO_4$. It sets because it turns back into gypsum.

Q17. Bleaching powder is made by passing chlorine gas over:

Answer: (a) dry slaked lime: $Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$

Q18. Riya got a turmeric (haldi) stain on her white shirt. When she scrubbed it with soap, the stain turned reddish-brown. After she rinsed it with plenty…

(i) Soap solution is basic, and turmeric is a natural indicator that turns reddish-brown in a base.
(ii) Rinsing washed the soap away, so the stain was no longer in a basic medium and turmeric showed its normal yellow colour.
(iii) Natural: litmus (or red cabbage juice). Synthetic: phenolphthalein (or methyl orange).

Q19. (a) A few drops of phenolphthalein are added to sodium hydroxide solution. Dilute hydrochloric acid is then added drop by drop. Describe the colour…

Answer: (a) The solution is pink at first, because phenolphthalein is pink in a base. As acid is added the pink fades, and it becomes colourless once all the base is neutralised. $NaOH + HCl \rightarrow NaCl + H_2O$
(b) It turns pink again, because the solution becomes basic again.
(c) Sodium chloride. Its solution is neutral (pH 7).

Q20. A calcium compound A reacts with dilute hydrochloric acid with effervescence. The gas given off puts out a burning candle. One of the products is…

Answer: A is calcium carbonate and the gas is carbon dioxide.
$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$